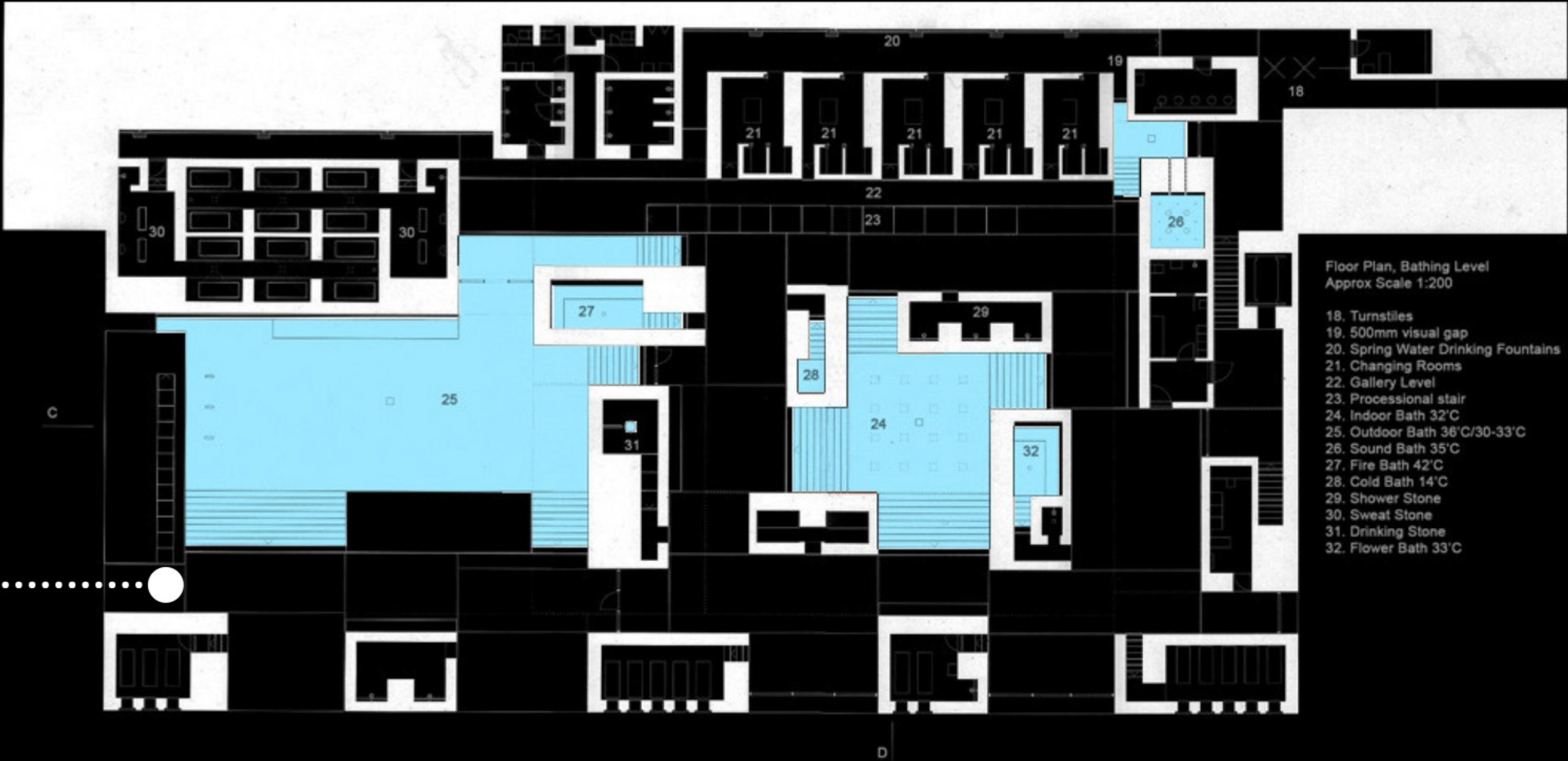
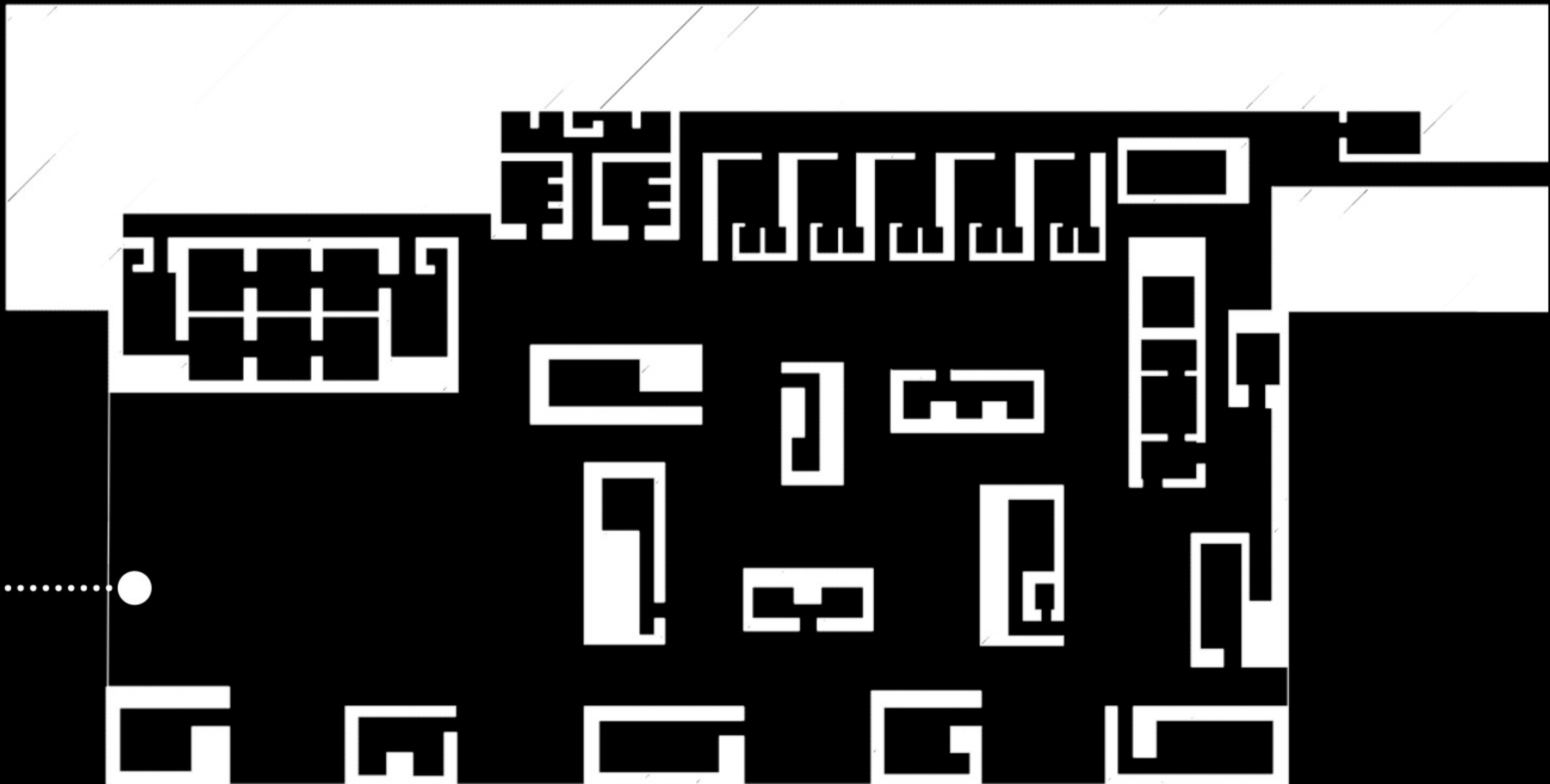
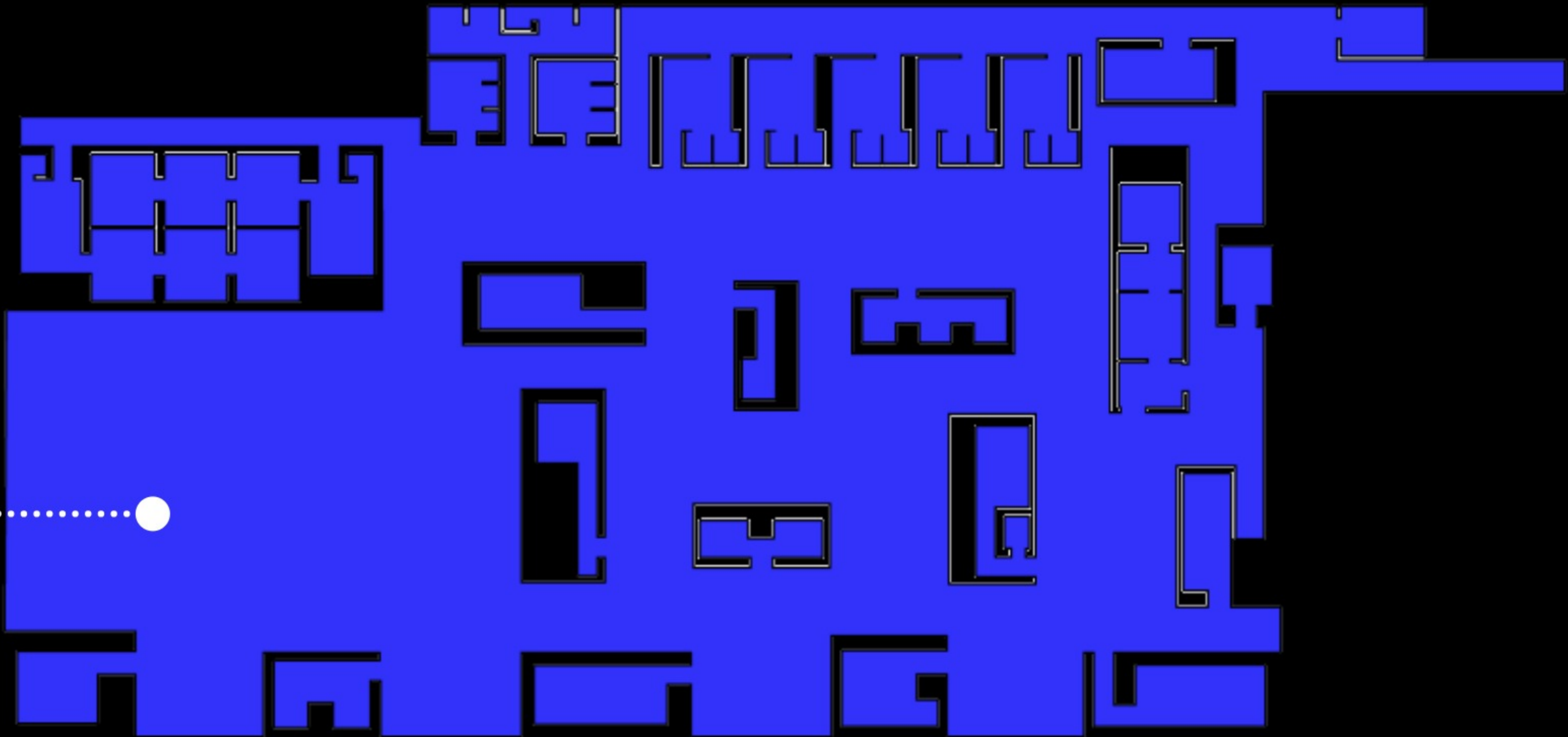




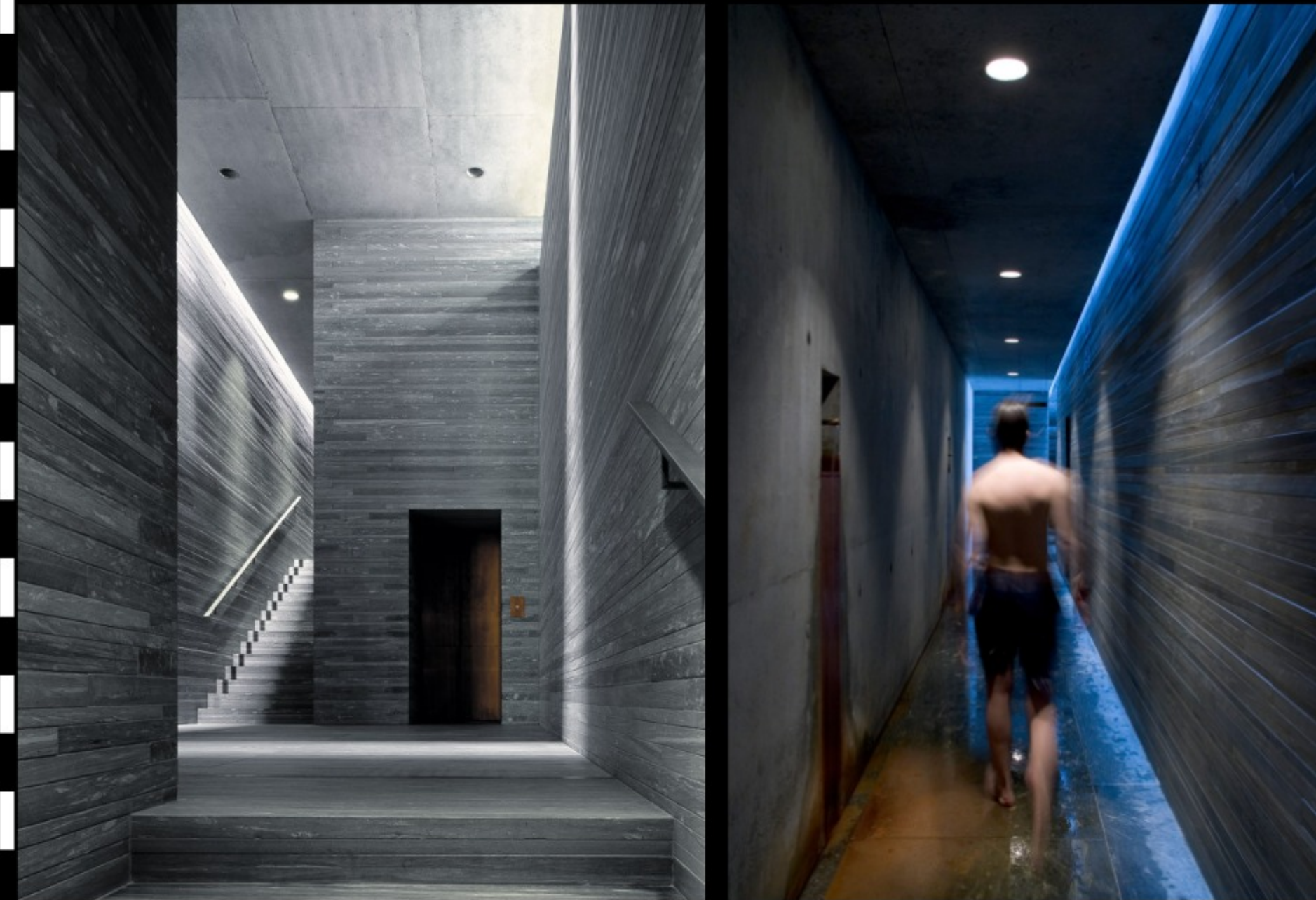
actual plan



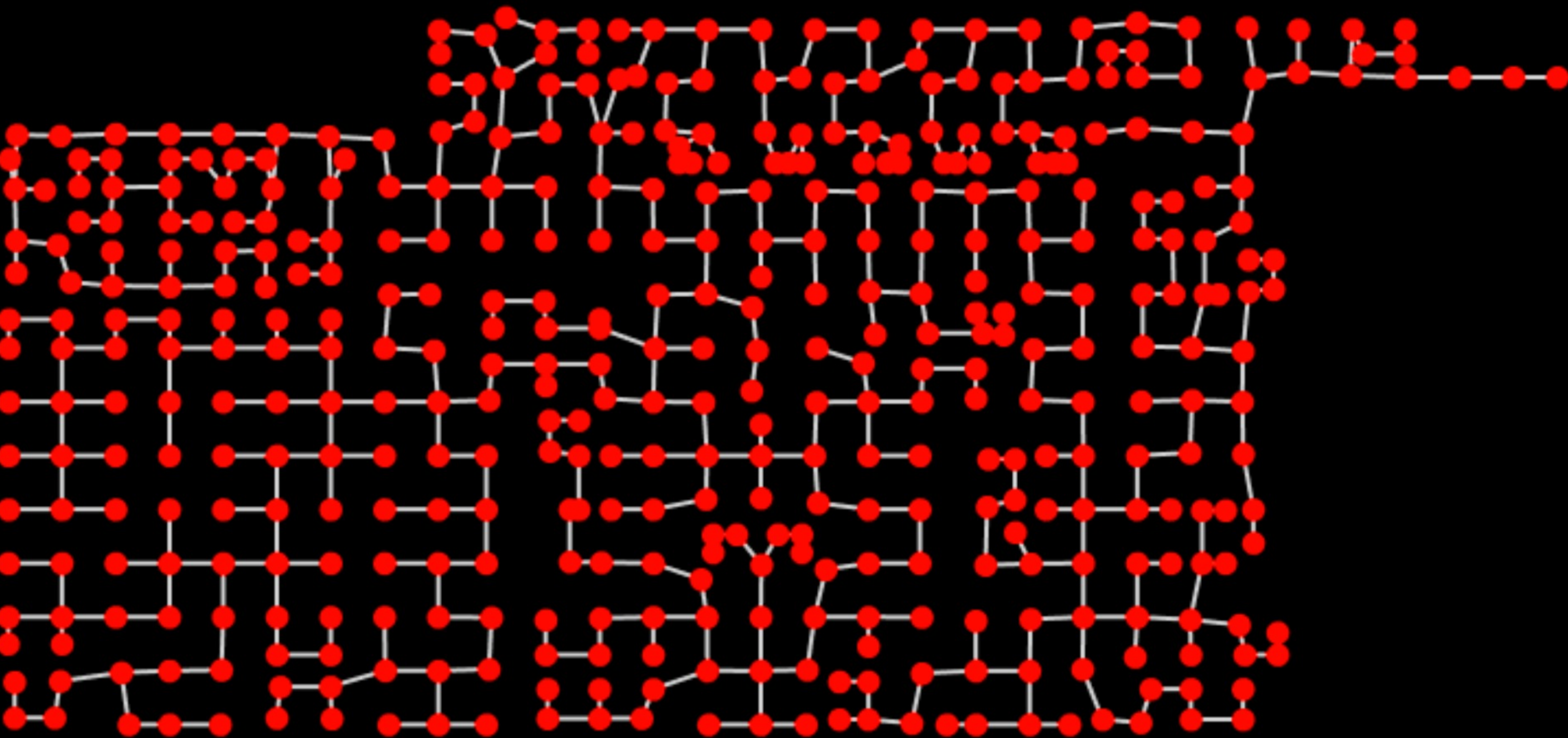
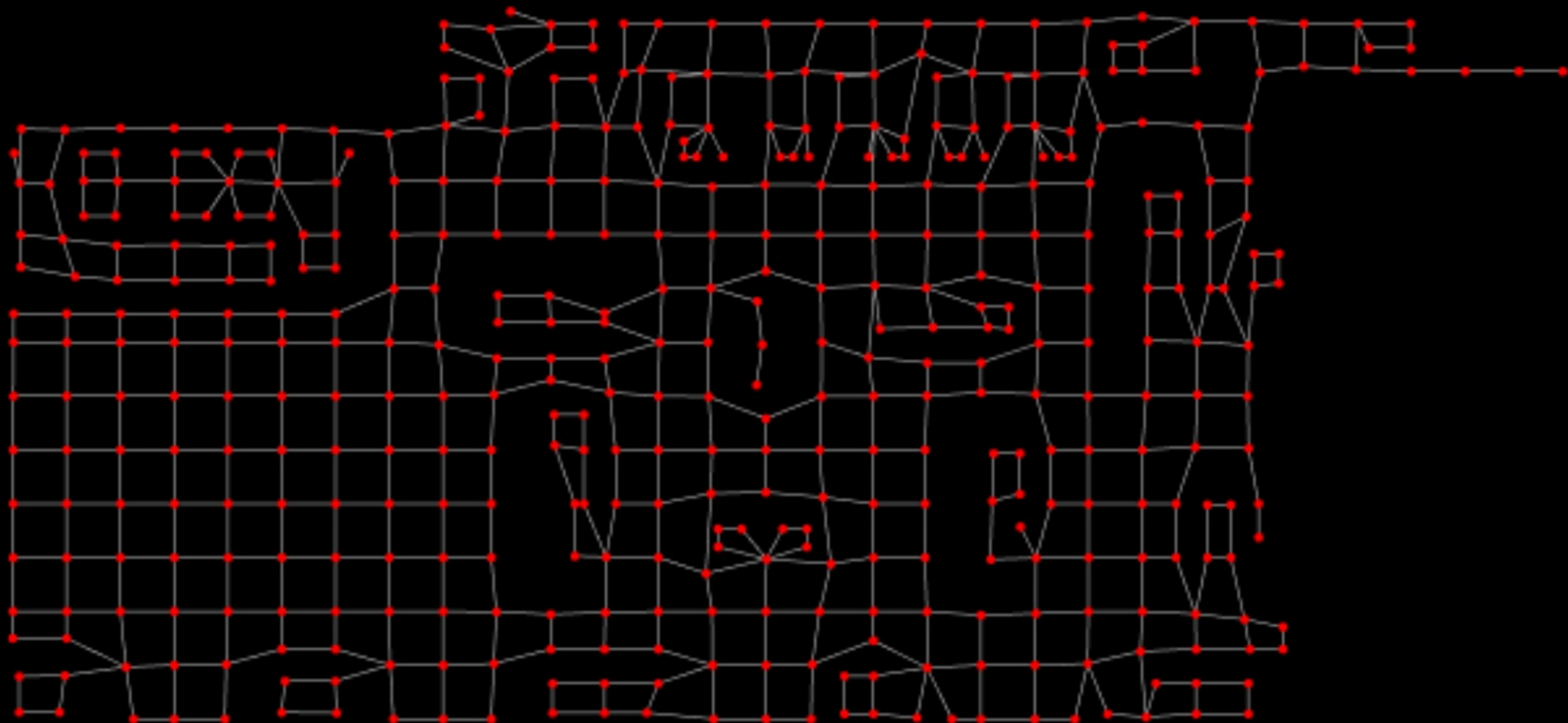
DXF



.brep - topology



Minimum Spanning Tree



Grid Overlay

Urange = 2 , 2

Vrange = 2 , 2

Vertices Analysis Graph

Density = 0.0071499

Diameter = 2: 41

Minimum Spanning Tree

Density = 0.0045146

Diameter = 70

Interpretation:

large, loosely connected networks
where most nodes are not directly
linked to each other.

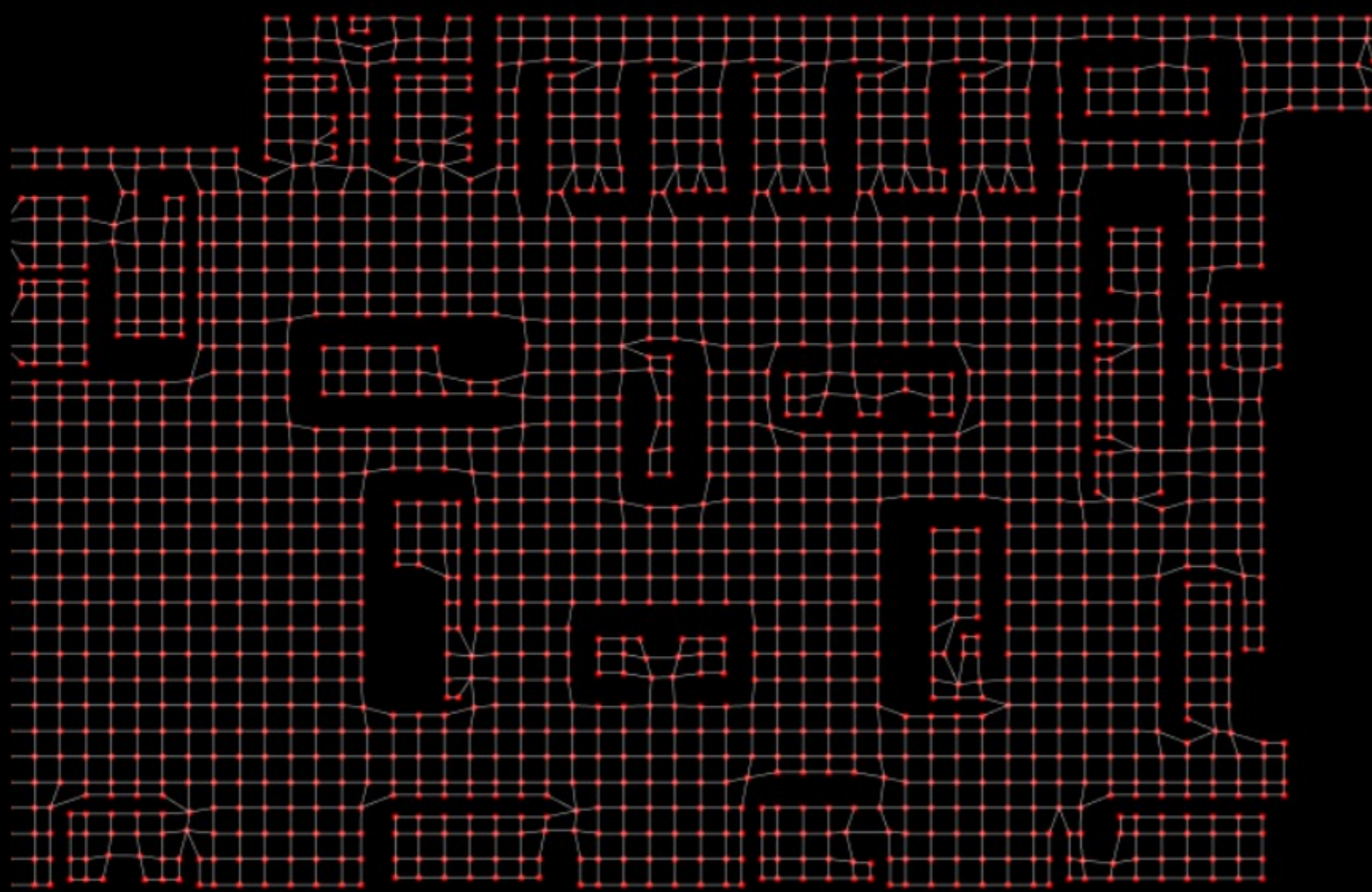
- hierarchical layouts
- deep & segregated buildings

Closeness Centrality



Grid Overlay

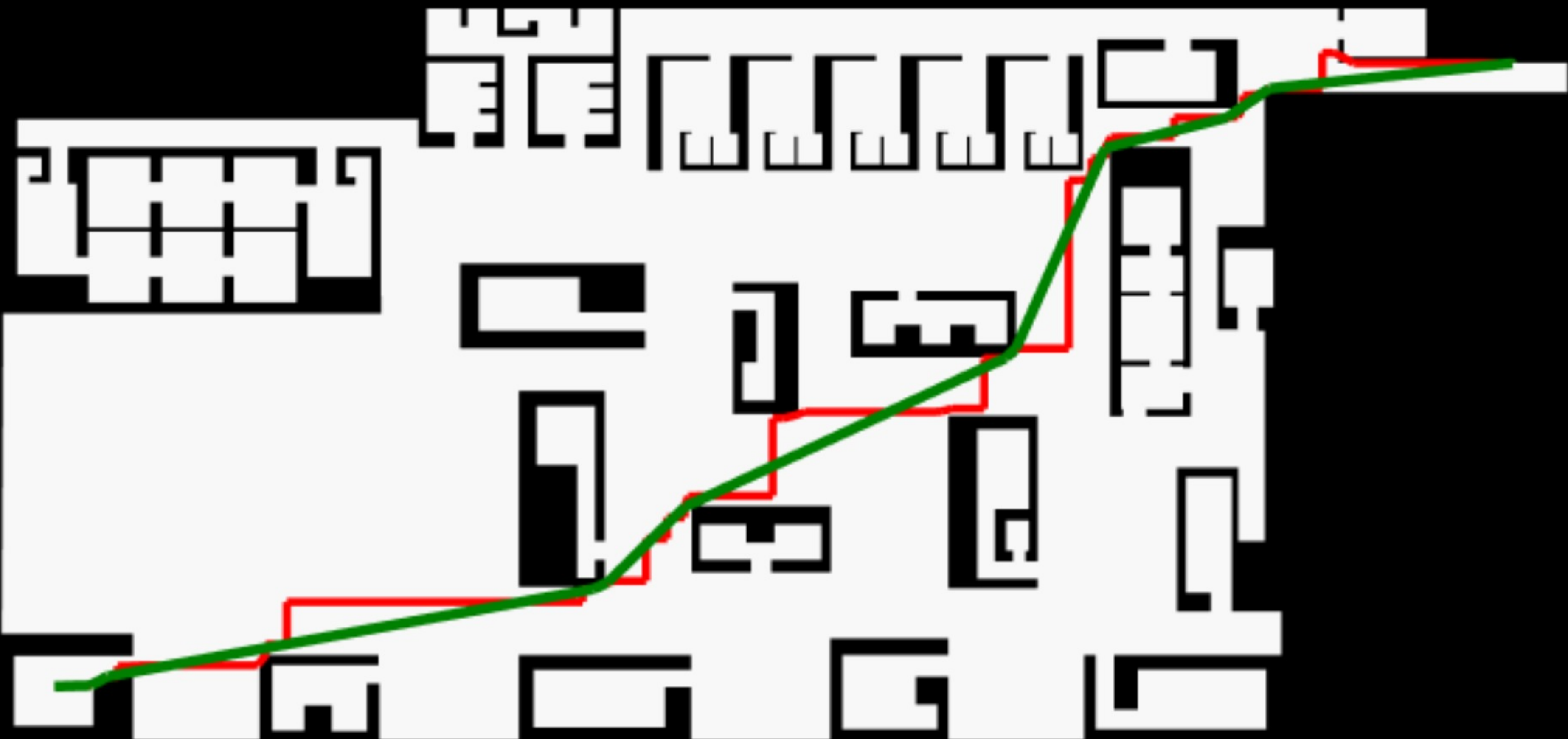
Urange = 0.8 , 0.8
Vrange = 0.8 , 0.8



Closeness Centrality

Min: 0.0159
Max: 0.0388
Mean: 0.0288
Spaces analysed: 2213

Shortest Path



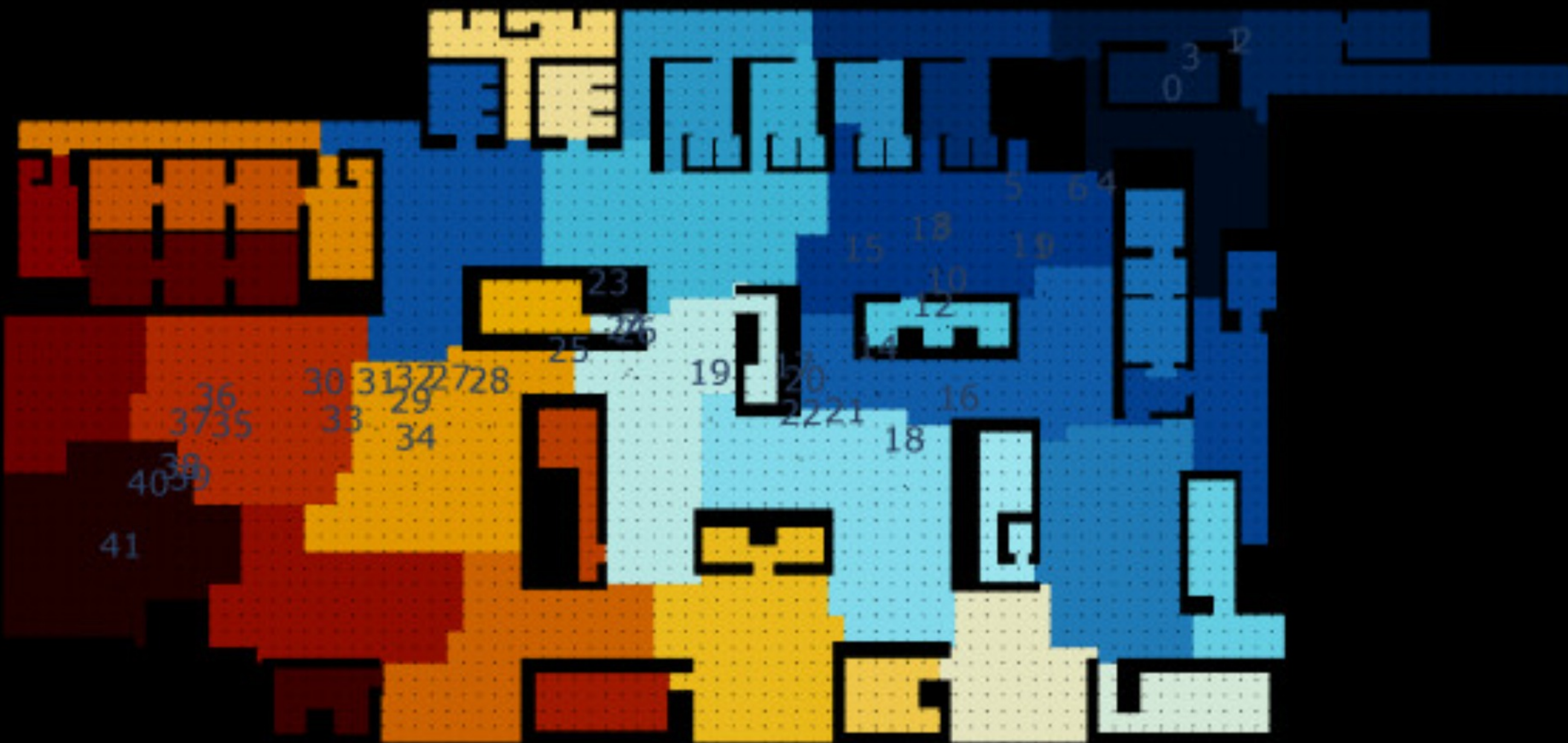
Shortest Path

Original SP Length: 77.23
Straightened SP Length: 63.66
Number of edges in path: 198
Number of turns: 197

Interpretation:

A flat, spread-out structure where most nodes are moderately connected to the rest.

Community detection



Grid Overlay

Urange = 0.8 , 0.8

Vrange = 0.8 , 0.8

Community detection

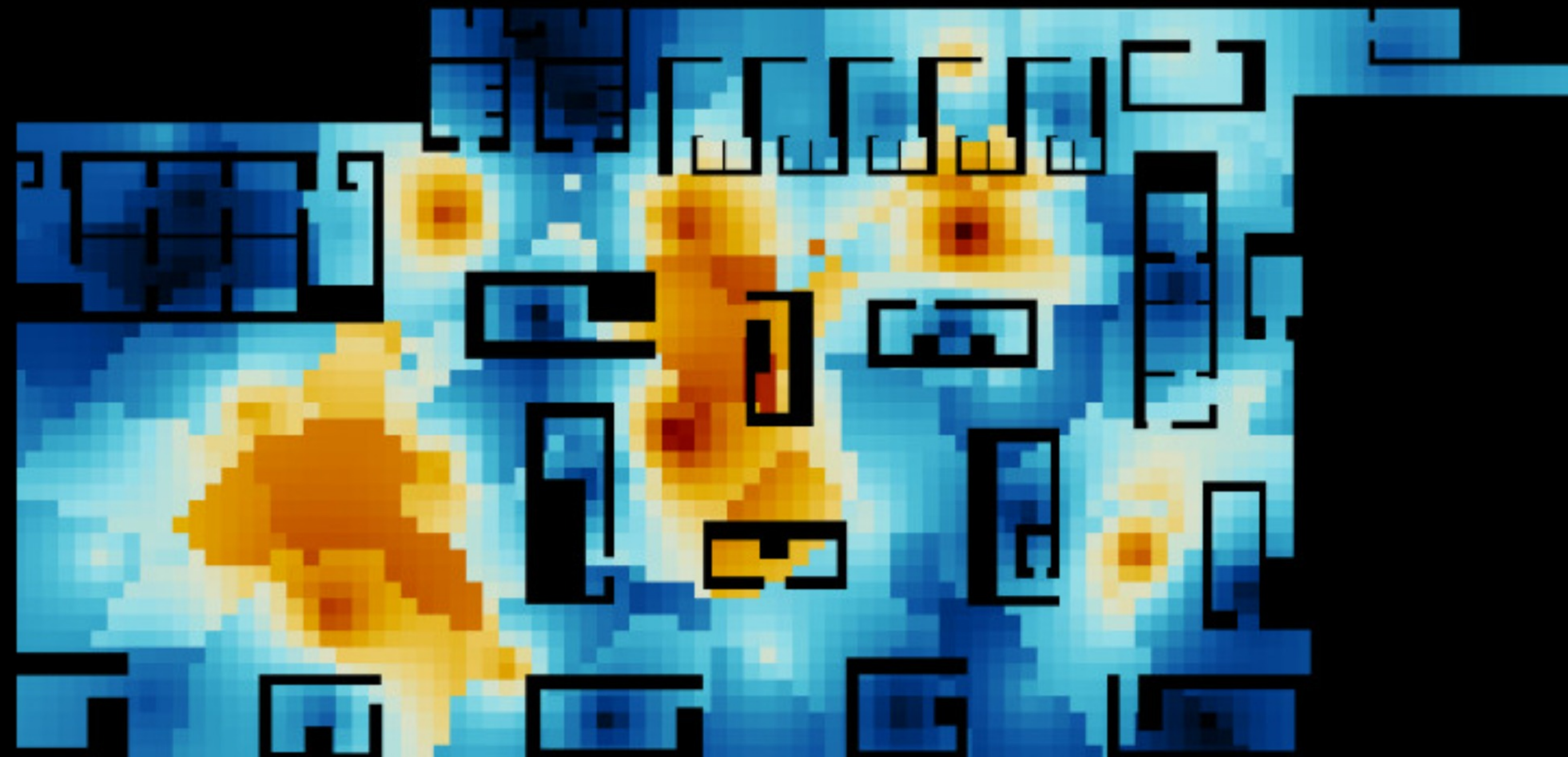
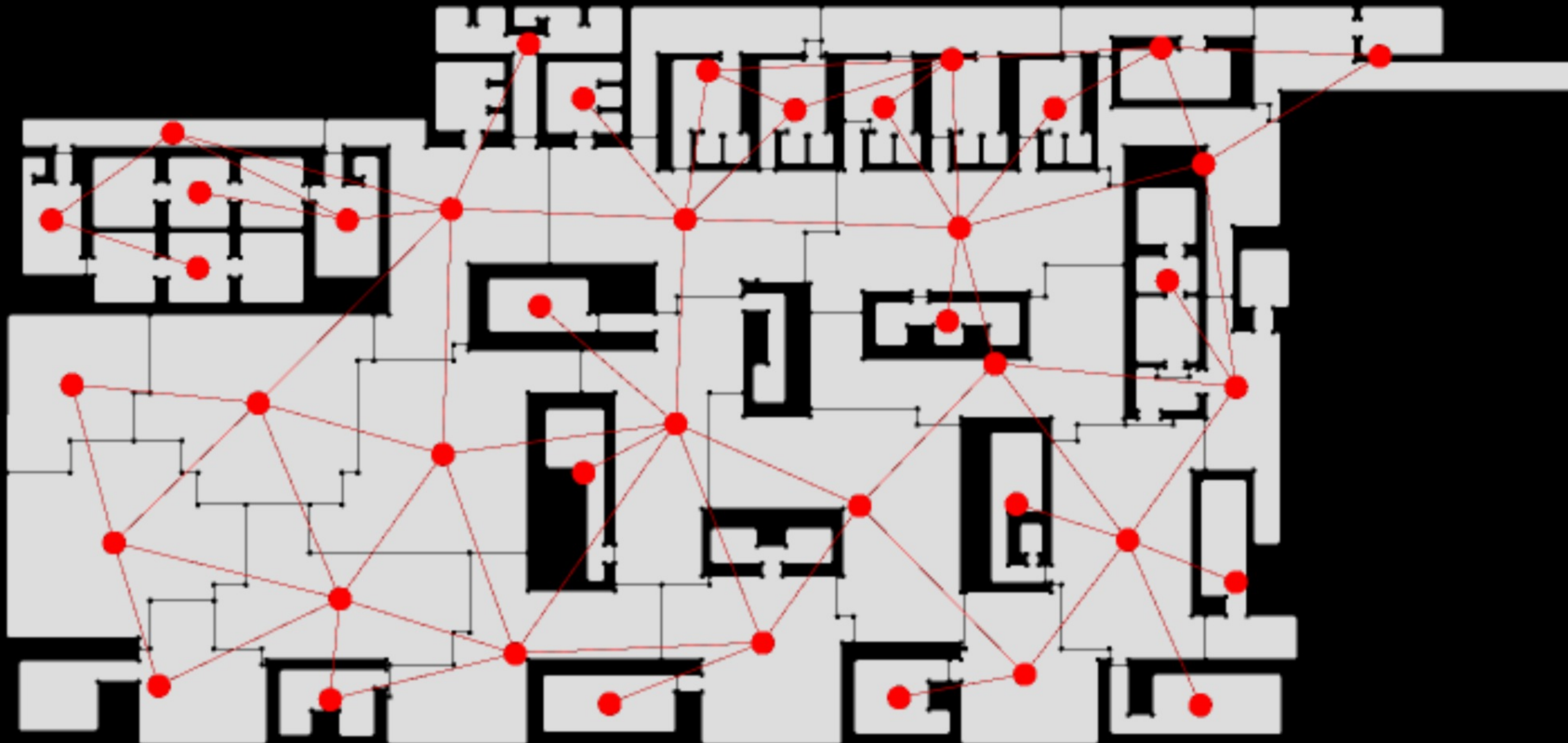
42 Community detected

- Community 0: 36 cells → ~23.0 m²
- Community 1: 77 cells → ~49.3 m²
- Community 2: 72 cells → ~46.1 m²
- Community 3: 114 cells → ~73.0 m²
- Community 4: 98 cells → ~62.7 m²
- Community 5: 161 cells → ~103.0 m²
- Community 6: 95 cells → ~60.8 m²
- Community 7: 172 cells → ~110.1 m²
- Community 8: 134 cells → ~85.8 m²
- Community 9: 63 cells → ~40.3 m²
- Community 10: 142 cells → ~90.9 m²
- Community 11: 37 cells → ~23.7 m²
- Community 12: 97 cells → ~62.1 m²
- Community 13: 40 cells → ~25.6 m²
- Community 14: 139 cells → ~89.0 m²
- Community 15: 38 cells → ~24.3 m²
- Community 16: 56 cells → ~35.8 m²
- Community 17: 174 cells → ~111.4 m²
- Community 18: 40 cells → ~25.6 m²
- Community 19: 152 cells → ~97.3 m²
- Community 20: 55 cells → ~35.2 m²
- Community 21: 84 cells → ~53.8 m²
- Community 22: 32 cells → ~20.5 m²
- Community 23: 72 cells → ~46.1 m²
- Community 24: 39 cells → ~25.0 m²
- Community 25: 141 cells → ~90.2 m²
- Community 26: 35 cells → ~22.4 m²
- Community 27: 160 cells → ~102.4 m²
- Community 28: 42 cells → ~26.9 m²
- Community 29: 60 cells → ~38.4 m²
- Community 30: 118 cells → ~75.5 m²
- Community 31: 70 cells → ~44.8 m²
- Community 32: 39 cells → ~25.0 m²
- Community 33: 150 cells → ~96.0 m²
- Community 34: 45 cells → ~28.8 m²
- Community 35: 115 cells → ~73.6 m²
- Community 36: 41 cells → ~26.2 m²
- Community 37: 77 cells → ~49.3 m²
- Community 38: 84 cells → ~53.8 m²
- Community 39: 35 cells → ~22.4 m²
- Community 40: 153 cells → ~97.9 m²
- Community 41: 107 cells → ~68.5 m²

Largest community (17) area: ~111.4 m²

Smallest community (22) area: ~20.5 m²

degree centralities



Grid Overlay

Urange = 0.8 , 0.8

Vrange = 0.8 , 0.8

degree centralities

Most connected:

7 links → score 0.1707

Least connected:

1 links → score 0.0244

Interpretation:

the orange spaces are the 7-link hubs, the blue spaces are your 1-link dead ends, and everything in between is the connective tissue of the building's spatial network.

most connected area are the large spaces

Betweenness Centrality



Grid Overlay

Urange = 0.8 , 0.8

Vrange = 0.8 , 0.8

Betweenness Centrality

Min: 0.0

Max: 0.156

Mean: 0.0159

Top 5 most traversed spaces:

Vertex at (27.35, 33.32) → BC: 0.156

Vertex at (26.55, 33.32) → BC: 0.153

Vertex at (25.75, 33.3) → BC: 0.1508

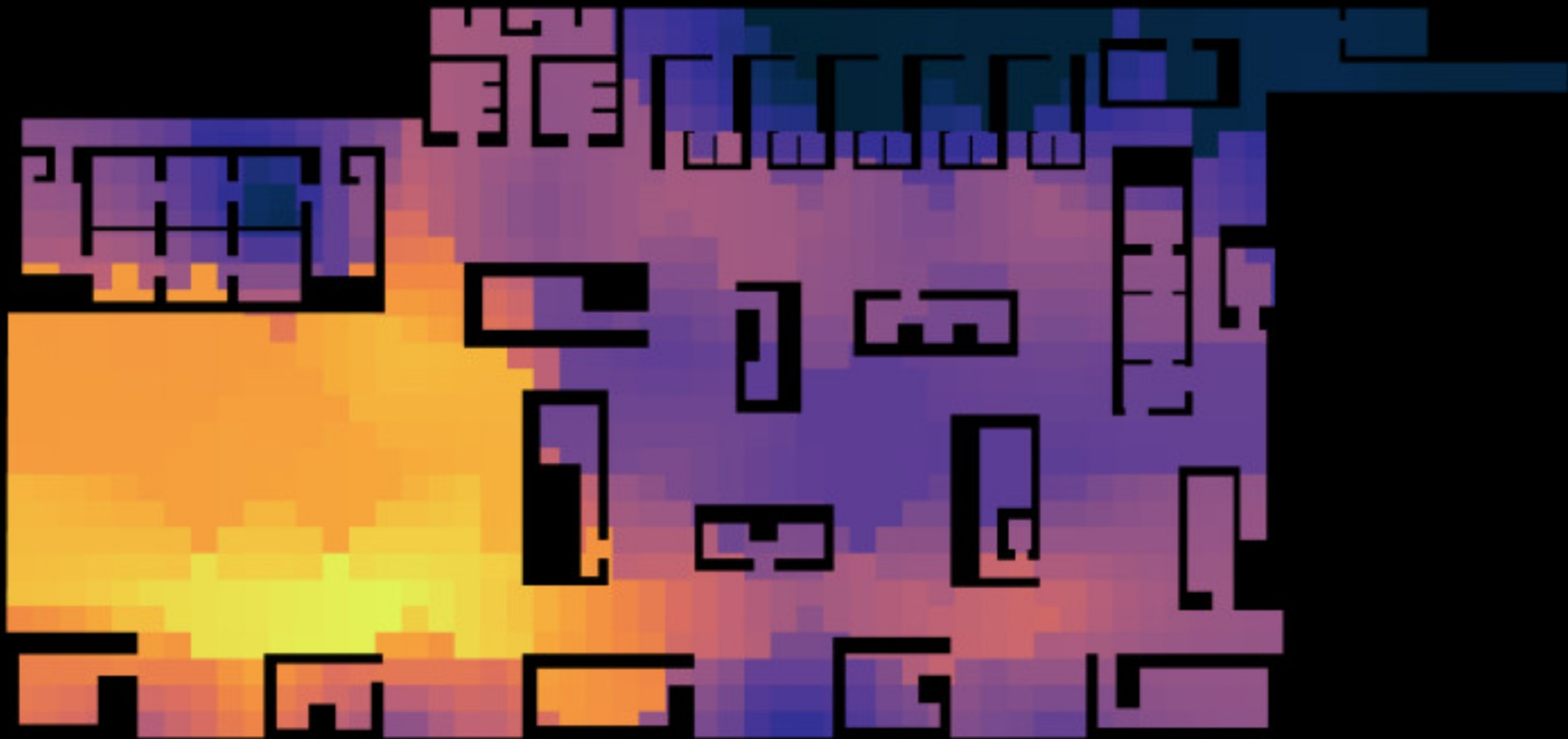
Vertex at (24.95, 33.31) → BC: 0.1052

Vertex at (24.15, 33.32) → BC: 0.0899

Interpretation:

The corridor is the structure where nearly all traffic concentrates. If that passage is blocked or removed, the network likely splits into disconnected parts. From a design perspective, this is both the most important and most vulnerable spot in the entire layout. which indicate to a spatial/architectural network.

Visibility Study



Grid Overlay

Urange = 0.6 , 0.6
Vrange = 0.6 , 0.6

Visibility Study

Valid visibility points: 36
Max visibility: 411
Min visibility: 18
Mean visibility: 200.39

Interpretation:

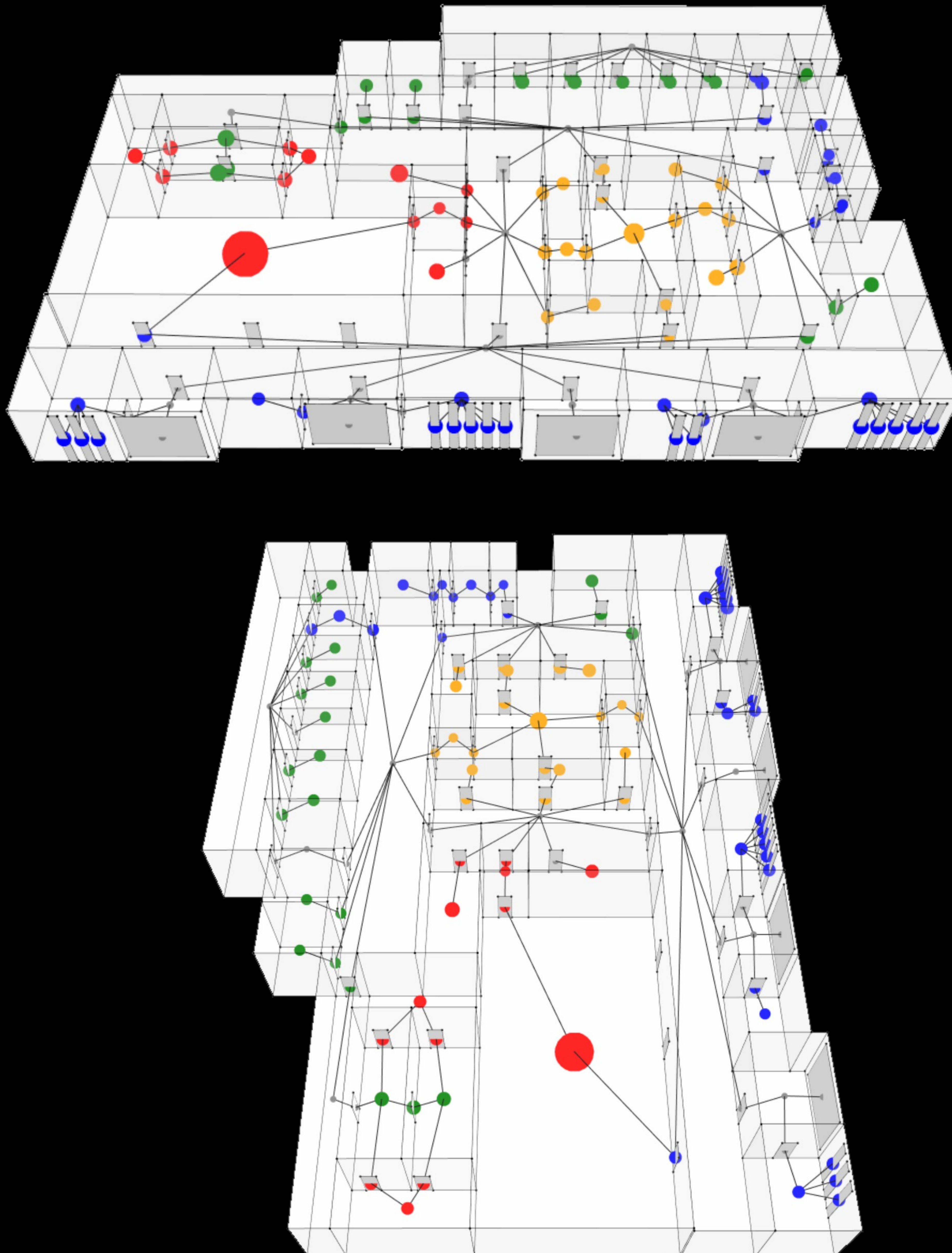
The visibility field is **highly uneven**, with strong concentration in one main zone.

The corridor acts as a **dominant spatial spine**, it organizes movement and perception across the layout.

- Efficient → clear navigation and focus
- Fragile → over-reliance on a single connector

spatial network is **centralized and hierarchical**, with one critical visibility corridor controlling both movement and connectivity,

Access/Circulation Graph



Grid Overlay

The logic follows Zumthor's design intention:

Therme Vals is not a building you navigate freely; it is a choreographed sequence of thermal sensations. By mapping adjacency as edges, the graph reveals which spaces are directly connected and which require passing through intermediate zones. The window connections show which spaces maintain a visual and atmospheric relationship with the exterior, distinguishing inward-focused thermal chambers from those oriented toward the landscape. Together, the graph makes the spatial and thermal logic of the building legible as a network structure.

Nodes represent each thermal space: hot pools, warm baths, cold rooms, neutral spaces, and corridors. Each node is colored by thermal category and sized proportionally to the room's surface area.

Edges represent two types of relationships:

The first is physical through door connection one can move between them. The second is exterior aperture connections; a room has a window opening to the outside, linking the enclosed thermal atmosphere to natural light and the landscape.